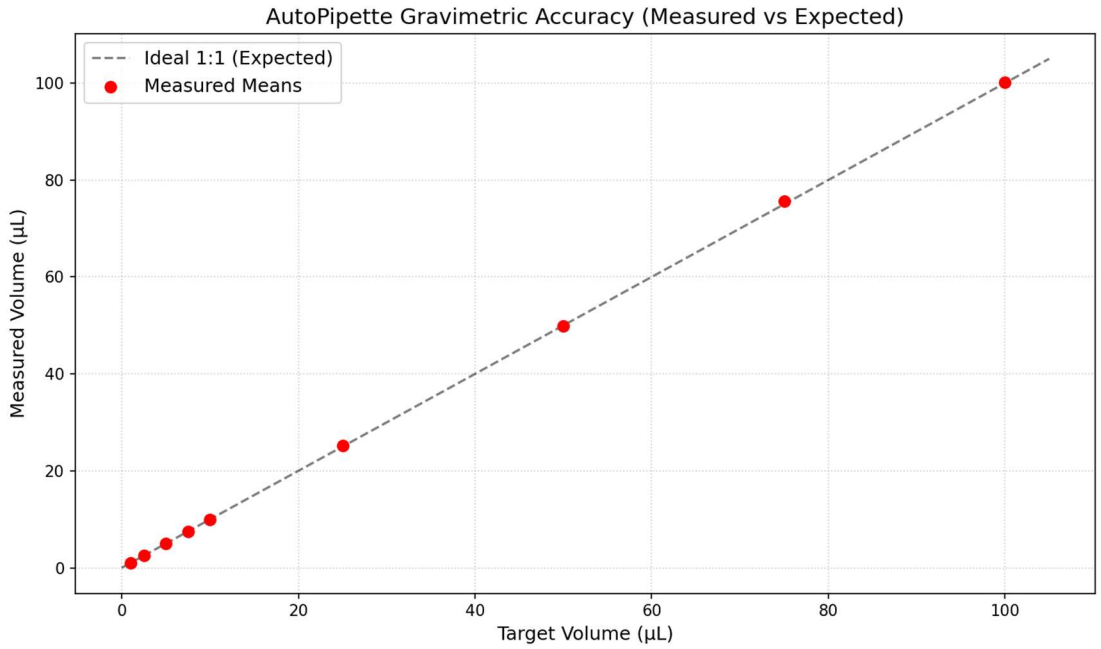
Pipette Calibration Report

1. Multi-Point Gravimetric Calibration

The following table shows the gravimetric calibration results.

Target (µL)	Mean (µL)	Accuracy (%)	CV (%)
1.0	1.000	+0.00%	10.00%*
2.5	2.600	+4.00%	3.85%
5.0	5.000	+0.00%	3.46%
7.5	7.533	+0.44%	1.53%
10.0	9.900	-1.00%	1.75%
25.0	25.200	+0.80%	0.40%
50.0	49.833	-0.33%	0.31%
75.0	75.600	+0.80%	0.13%
100.0	100.100	+0.10%	0.10%

[!NOTE] *The 10.00% CV at 1.0 µL is artificially inflated by the 0.1 mg scale resolution limit, meaning the true precision is better than reported.*



Gravimetric Accuracy Plot

2. Spectrophotometric Dye-Dilution Analysis

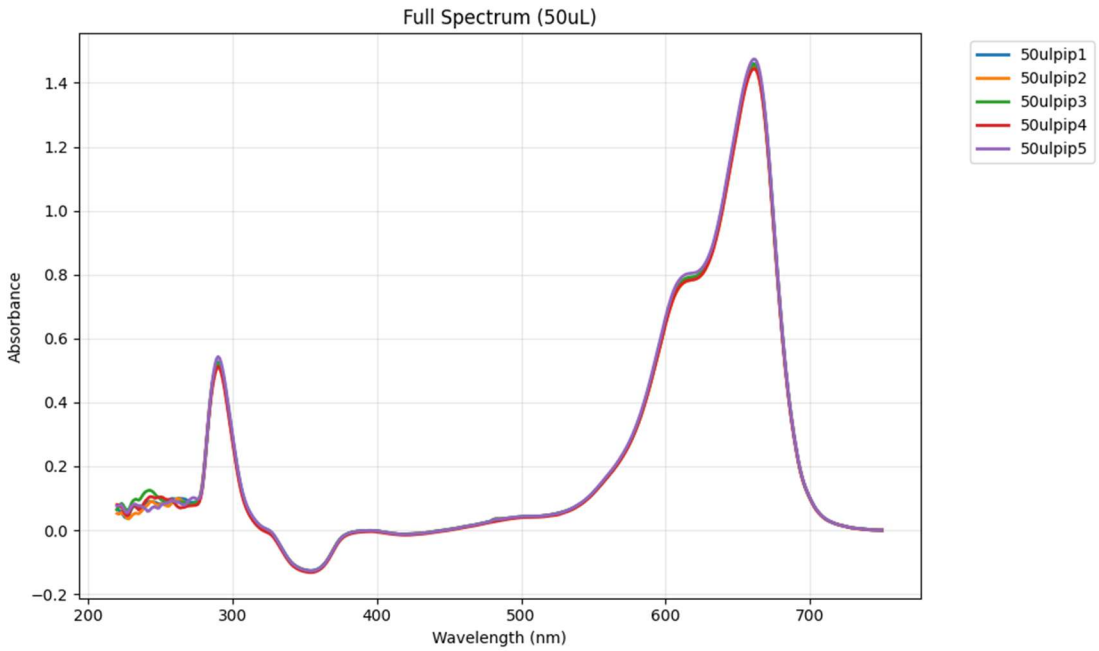
Using dye-dilution and measuring absorbance at 665 nm, the %CV in absorbance acts as a direct proxy for the %CV in volume.

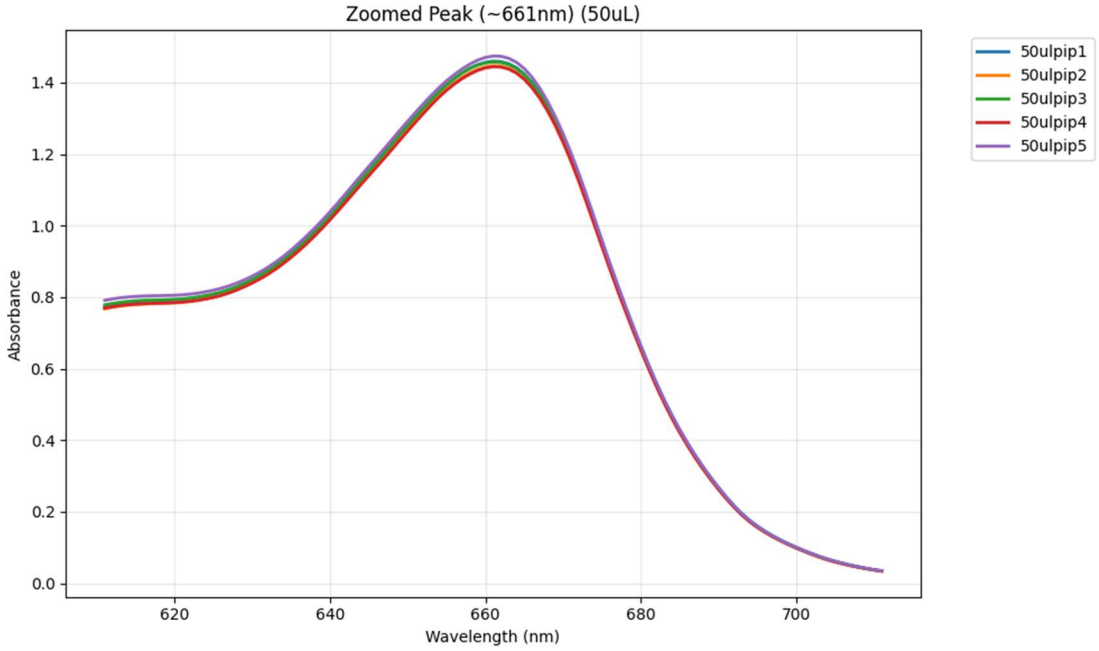
50 µL (Full) Aliquots Analysis

At a 50 µL nominal volume, the AutoPipette performs exceptionally well, rivaling commercial pipettes like Eppendorf or Gilson (which typically spec <1% CV at 50 µL).

Metric	CV%	→ Volume SD (at 50 µL)	→ Volume Variance
Instrument (Spec) repeatability	0.37%	0.19 µL	0.034 μL^2
Pipette Variation (Real Spread)	**0.54%**	**0.27 µL**	**0.072 μL^2**
Overall (all valid readings)	0.59%	0.29 µL	0.086 μL^2

Conclusion: The actual pipette-to-pipette variation is about **±0.27 µL**, meaning the AutoPipette reliably delivers **50 ± 0.27 µL**.





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10 µL Aliquots Analysis

At 10 µL, the AutoPipette's variance and precision were tested independently.

Source	CV%	→ SD in µL	→ Variance
Instrument (Spec) Noise	0.07%	0.007 µL	0.00005 μL^2
AutoPipette	**2.09%**	**0.21 µL**	**0.043 μL^2**

[!NOTE] **Precision Analysis (Prior to Firmware Fixes)** The AutoPipette's raw draw-to-draw CV at 10 µL was measured at 2.09%. This precision measurement highlights the baseline mechanical repeatability prior to the implementation of the new Multipoint Calibration firmware.

